

Programme	B. Sc. Statistics				
Course Code	STA4CJ203(P)				
Course Title	Applied Statistics Time Series, Index Numbers & official statistics				
Type of Course	Major				
Semester	IV				
Academic Level	200-299				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	HSE level Mathematics/Statistics courses				
Course Summary Objective	Make students to apply statistical models in time series data. Importance of various indices and vital rates.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Describing TS components forces affecting data.	U	C	Instructor-created exams / Quiz
CO2	Interpreting and computing trend and SI	Ap	P	Practical Assignment / Observation of Practical Skills
CO3	Relate IN for economic policy formulation	Ap	P	Seminar Presentation / Group Tutorial Work
CO4	Computation of various IN	U	C	Instructor-created exams / Home Assignments
CO5	Summarise Fertility mortality rates	Ap	P	One Minute Reflection Writing assignments
CO6	Construct Life tables	Ap	P	Viva Voce
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (45 +30)	Marks (70)
I	Time Series		10	20
	1	Time series definition and Components of time series.		
	2	Additive and Multiplicative models		
	3	Measurement of secular trend Free Hand/Graphical method		
	4	Method of Semi Averages		
	5	Method of moving averages		
	6	Method of least squares (linear, quadratic and exponential).		
	Sections from References:			
II	Measurement of Seasonal Variation		10	15
	7	Simple average method.		
	8	Ratio to trend Method		
	9	Ratio to moving average		
	10	Method-Link relative method		
	Sections from References:			
III	Index Numbers		14	20
	11	Classification of Index Numbers		
	12	Methods of constructing Index Numbers		
	13	Unweighted Index Numbers, Weighted Index Numbers		
	14	Laspeyre's, Paasche's, Marshal-Edgeworth, Fisher's, Dorbish Bowleys, Kellys)-		
	15	Quantity Index Numbers-Fixed base and chain base Index. Numbers		
	16	Different tests of a good Index numbers: - Unit test, Time Reversal Test-Factor Reversal Test- Circular test.		
	17	Fishers Ideal Index Number		
	18	Cost of Living Index Numbers-Consumer Price Index Numbers-		
	19	Family Budget enquiry		
	Sections from References:			
IV	Vital Statistics		11	15
	20	Sources of Vital Statistics (SRS, CRS),		
	21	Fertility rate- CBR, ASFR, TFR, GFR,		
	22	Mortality rate- CDR, ASDR, SDR, IMR,		
	23	Population growth- NRR and GRR (definitions only).		
	24	Construction of simple life tables		
	Sections from References:			
V	Open Ended Module: Practical Applications, Case Study		30	
	1	Visit of Government Organizations NSSO, DES Case study using secondary data available from government publications of Module I, II, III & IV Presentation of collected data.		
	Sections from References:			